Module Details

Module Code	**UFCFK1-15-0**
Module Title	**Fundamentals of Data Science**
Module Tutors	**Saurav Gautam**
Year	**2025-2026**
Component/Element Number	**PSA/Bi-weekly assignment/Regular**
Weighting	**10%**

Dates

Submission Date	**17-April-2026**
Submission Place	**Backboard**
Submission Time	**23:59**
Submission Notes	**Submit Gitlab URL**

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1. Write a program to create a function that accepts **two integers** and displays their **sum, difference, product, quotient** values. **Marks** [10 marks]

```
def calculate(a, b): # function to perform arithmetic operations on two
numbers

   sum_result = a + b

   difference_result = a - b

   product_result = a * b

    quotient_result = a / b if b != 0 else None  # Handle division by zero
case

   return sum_result, difference_result, product_result, quotient_result

# Main program

num1 = int(input("Enter the first integer: "))

num2 = int(input("Enter the second integer: "))

sum_, difference, product, quotient = calculate(num1, num2)  # calling the
calculate function with user inputs

# Displaying the results

print(f"Sum: {sum_}")

print(f"Difference: {difference}")

print(f"Product: {product}")

print(f"Quotient: {quotient if quotient is not None else 'Division by zero
is undefined.'}")
```

Output:

```
Enter the first integer: 10
Enter the second integer: 20

RESULTS

Addition:
 10 + 20 = 30

Multiplication:
 10 * 20 = 200

Division:
 10 / 20 = 0.50

Floor Division:
 10 // 20 = 0

Exponentiation:
 10 ** 20 = 100000000000000000000
```

2. Write a program to implement separate functions for the following mathematical operations. Each function should take **two user inputs as parameters**, operate, and return the result. Display the results clearly with **proper formatting. Marks [**2*5 = 10]

   I. Addition

   II. Multiplication

   III. Division

   IV. Floor Division

   V. Exponentiation

```
def addition(a, b): # function to perform addition of two numbers

   return a + b

def multiplication(a, b): # function to perform multiplication of two
numbers

   return a * b
```

```
def division(a, b): # function to perform division of two numbers, with handling for division by zero

   if b != 0: return a / b

   else: return None

def floor_division(a, b): # function to perform floor division of two numbers, with handling for division by zero

   if b != 0: return a // b

   else: return None

def exponentiation(a, b): # function to perform exponentiation of two numbers

   return a ** b

# Main program

num1 = int(input("Enter the first integer: "))

num2 = int(input("Enter the second integer: "))



# Storing results of operations in a dictionary for cleaner output formatting

operations = {

   "Addition": addition(num1, num2),

   "Multiplication": multiplication(num1, num2),

   "Division": division(num1, num2),  # FIXED

   "Floor Division": floor_division(num1, num2),

   "Exponentiation": exponentiation(num1, num2)
```

```
}

symbols = {

    "Addition": "+",

    "Multiplication": "*",

    "Division": "/",

    "Floor Division": "//",

    "Exponentiation": "**"

}

# Displaying results in a clean format, with special handling for division
by zero cases

print("\nRESULTS")

for operation, result in operations.items():

    print()

    if result is None:

        print(f"{operation}:\n Undefined (division by zero).")

    else:

        if operation == "Division":

                print(f"{operation}:\n {num1} {symbols[operation]} {num2} =
{result:.2f}")

        else:

                print(f"{operation}:\n {num1} {symbols[operation]} {num2} =
{result}")
```

Output:

```
Enter the first integer: 10
Enter the second integer: 2

RESULTS

Addition:
 10 + 2 = 12

Multiplication:
 10 * 2 = 20

Division:
 10 / 2 = 5.00

Floor Division:
 10 // 2 = 5

Exponentiation:
 10 ** 2 = 100
```

3. Write a program to create a function that accepts one string as input and checks whether it is **Armstrong** or not. Display the output neatly. **Marks** [10]

```
def is_armstrong(str_num): # function to check if a number is an Armstrong number

    res = int(str_num)

    sum = 0

    power = len(str_num)  # The power is determined by the number of digits in the number

    for x in str_num:

        sum += int(x) ** power  # Summing the digits raised to the power



    return res == sum  # An Armstrong number is one where the sum of its own digits each raised to the power of the number of digits equals the number itself
```

```python
while True: # validating input to ensure it's an integer
    try:
        num = int(input("Enter a number: "))
        break
    except ValueError:
        print("Invalid input! Please enter a valid integer.")



if(is_armstrong(str(num))):
    print(f"{num} is an Armstrong number.")
else:
    print(f"{num} is not an Armstrong number.")
print()
```

Output:

```
7 -- /home/erza/python/assignment2/ar
Enter a number: 121
121 is not an Armstrong number.

erza@erza-HP-Pavilion-Laptop-14-dv2xx
adapter/../../debugpy/launcher 60285
Enter a number: 153
153 is an Armstrong number.
```

4. Write a function that takes a list of numbers as input and returns a dictionary showing how many times each number occurs in the list. Test this function with at least **three different lists**. **Marks** [10]

```
def count_occurance(num_list): # function to count the occurances of each
number in the list
   occurances = {}
   for num in num_list:
       if num in occurances:
           occurances[num] += 1
       else:
           occurances[num] = 1
   return occurances

# Test cases
test_list1 = [1, 2, 2, 3, 4, 4, 4, 5]
test_list2 = [10, 20, 20, 30, 40, 40, 40, 50]
test_list3 = [5, 5, 5, 5, 5]
test_list4 = [1, 2, 3, 4, 5]  # a test case with unique numbers
counts1 = count_occurance(test_list1)
counts2 = count_occurance(test_list2)
counts3 = count_occurance(test_list3)
counts4 = count_occurance(test_list4)

print("Test List 1:", test_list1)
print("Occurrences:", counts1)
print()

print("Test List 2:", test_list2)
print("Occurrences:", counts2)
print()

print("Test List 3:", test_list3)
print("Occurrences:", counts3)
print()

print("Test List 4:", test_list4)
print("Occurrences:", counts4)
print()
```

```
while True: # validating input to ensure it's a list of integers
   try:
         user_input = input("Enter a list of numbers separated by commas
(e.g., 1,2,3): ")
       user_list = [int(x.strip()) for x in user_input.split(",")]
       break
   except ValueError:
           print("Invalid input! Please enter a valid list of integers
separated by commas.")

counts_user = count_occurance(user_list)
print("User List:", user_list)
print("Occurrences:", counts_user)
```

Output:

```
erza@erza-HP-Pavilion-Laptop-14-dv2xxx:~/python/assignment2$ /usr/bin/python3 /home/erza/python/assignment2/number_occurance
Test List 1: [1, 2, 2, 3, 4, 4, 4, 5]
Occurrences: {1: 1, 2: 2, 3: 1, 4: 3, 5: 1}

Test List 2: [10, 20, 20, 30, 40, 40, 40, 50]
Occurrences: {10: 1, 20: 2, 30: 1, 40: 3, 50: 1}

Test List 3: [5, 5, 5, 5, 5]
Occurrences: {5: 5}

Test List 4: [1, 2, 3, 4, 5]
Occurrences: {1: 1, 2: 1, 3: 1, 4: 1, 5: 1}

Enter a list of numbers separated by commas (e.g., 1,2,3): 1,2,3,5,3,1,1,2,2,1,2,3,2,2,3,4,5,6,7,8
User List: [1, 2, 3, 5, 3, 1, 1, 2, 2, 1, 2, 3, 2, 2, 3, 4, 5, 6, 7, 8]
Occurrences: {1: 4, 2: 6, 3: 4, 5: 2, 4: 1, 6: 1, 7: 1, 8: 1}
```

5. Write a function that accepts a list of student names and returns the names sorted in **reverse alphabetical order**. **Marks** [10]

```
def reverse_sort_student_names(names):
   return sorted(names, reverse=True)

names = []
while True:

   name = input("Enter student name: ")
   names.append(name)

   if input("Do you want to enter another name? (y/n): ").lower() != "y":
       break
```

```
sorted_names = reverse_sort_student_names(names)
print("Student names in reverse alphabetical order:", sorted_names)
```

Output:

```
erza@erza-HP-Pavilion-Laptop-14-dv2xxx:~/python/assignment2$ /usr/bin/python3 /home
Enter student name: ram
Do you want to enter another name? (y/n): y
Enter student name: zebra
Do you want to enter another name? (y/n): y
Enter student name: ads
Do you want to enter another name? (y/n): y
Enter student name: hari
Do you want to enter another name? (y/n): n
Student names in reverse alphabetical order: ['zebra', 'ram', 'hari', 'ads']
erza@erza-HP-Pavilion-Laptop-14-dv2xxx:~/python/assignment2$
```

6. Write a function that accepts two parameters: a list of countries and the country to be searched. The function should return the **index of the country** if found. Otherwise, it should return the message **"Not Found in List"**. **Marks** [10]

```
def find_country(countries , country_to_find):
   if country_to_find in countries:
       return countries.index(country_to_find)
   else:
       return "Not Found in List"

countries = []
while True:
   country = input("Enter a country: ")
   countries.append(country)

    if input("Do you want to enter another country? (y/n): ").lower() !=
"y":
       break

country_to_find = input("Enter the country to search for: ")
result = find_country(countries, country_to_find)
print(result)
print(f"Countries: {countries}")
```

Output:

```
erza@erza-HP-Pavilion-Laptop-14-dv2xxx:~/python/assignment2$ /usr/bin/python3 /home
Enter a country: nepal
Do you want to enter another country? (y/n): y
Enter a country: india
Do you want to enter another country? (y/n): y
Enter a country: china
Do you want to enter another country? (y/n): n
Enter the country to search for: nepal
Index: 0
Countries: ['nepal', 'india', 'china']
erza@erza-HP-Pavilion-Laptop-14-dv2xxx:~/python/assignment2$
```

7. Write a function **key_value(sentence)** that takes a sentence as input and returns a dictionary showing how many times each character (excluding spaces) occurs. **Marks [**10]

```python
def key_value(sentence):
    character_count = {}
    for char in sentence:
        if char != " ":
            if char in character_count:
                character_count[char] += 1
            else:
                character_count[char] = 1


    return character_count


sentence = input("Enter a Sentence: ")
result = key_value(sentence)
print(result)
```

Output:

```
erza@erza-HP-Pavilion-Laptop-14-dv2xxx:~/python/assignment2$ /usr/bin/python3 /home/erza/python/assignment2/count_
Enter a Sentence: hi how are you. i m fine
{'h': 2, 'i': 3, 'o': 2, 'w': 1, 'a': 1, 'r': 1, 'e': 2, 'y': 1, 'u': 1, '.': 1, 'm': 1, 'f': 1, 'n': 1}
erza@erza-HP-Pavilion-Laptop-14-dv2xxx:~/python/assignment2$
```

8. Write a program to accept a list of numbers from the user and return a new list with **all duplicates removed and the numbers sorted** in ascending order. **Marks [**10]

```
def remove_duplicates_and_sort(numbers):
    unique_numbers = set(numbers)  # Remove duplicates using a set
    sorted_numbers = sorted(unique_numbers)  # Sort the unique numbers
    return sorted_numbers

while True:
    try:
        user_input = input("Enter a list of numbers separated by commas: ")
        number_list = [int(num.strip()) for num in user_input.split(",")]
        result = remove_duplicates_and_sort(number_list)
        print("Sorted list with duplicates removed:", result)
        break
    except ValueError:
         print("Invalid input. Please enter a list of numbers separated by
commas.")
```

Output:

```
Enter a list of numbers separated by commas: 1,2,3,2,1,2,3,3,5,5,7,1,3,7,8,9,0
Sorted list with duplicates removed: [0, 1, 2, 3, 5, 7, 8, 9]
```

9. Write a program to ask details of **3 movies** (title, director, release year, rating). Store them in a dictionary and display the information in a well-formatted manner. **Marks [**10]

```
def get_info():
    movie_info = []
    ask_details = ["title", "director", "release year", "rating"]

    for i in range(3):
        movie = {}
        print(f"\nEnter the details of movie {i+1}:")
        for detail in ask_details:
            while True:
                try:
                    if detail == "release year":
```

```
                        value = input("Enter the release year: ").strip()
                        if value == "":
                            print("Release year cannot be empty.")
                            continue
                        value = int(value)

                    elif detail == "rating":
                        value = input("Enter the rating: ").strip()
                        if value == "":
                            print("Rating cannot be empty.")
                            continue
                        value = float(value)
                        if value < 0 or value > 10:
                            print("Rating must be between 0 and 10.")
                            continue

                    else:
                        value = input(f"Enter the {detail}: ").strip()
                        if value == "":
                                    print(f"{detail.capitalize()} cannot be
empty.")
                            continue

                    movie[detail] = value
                    break

                except ValueError:
                    print("Invalid input. Please try again.")

        movie_info.append(movie)

    return movie_info

info = get_info()

print("\nMovie Details:\n")
for i in range(len(info)):
    print(f"Movie {i+1}:")
    for key, value in info[i].items():
        print(f"{key.capitalize()}: {value}")
```

Output:

```
Enter the details of movie 1:
Enter the title: ram
Enter the director: ram123
Enter the release year: 1999
Enter the rating: 5

Enter the details of movie 2:
Enter the title: hari
Enter the director: hari bahadur
Enter the release year: 2020
Enter the rating: 6

Enter the details of movie 3:
Enter the title: ads
Enter the director: ads ads
Enter the release year: 2028
Enter the rating: 10

Movie Details:

Movie 1:
Title: ram
Director: ram123
Release year: 1999
Rating: 5.0

Movie 2:
Title: hari
Director: hari bahadur
Release year: 2020
Rating: 6.0

Movie 3:
Title: ads
Director: ads ads
Release year: 2028
Rating: 10.0
```

10. Create a program to design a **dice guessing game** with the following specifications: **Marks [**10]

    I. The dice has values from 1 to 6.

    II. The program should generate a random value for the dice roll.

    III. The player should guess the dice value.

    IV. If the guess is correct, show a **smiling face**.

    V. If the guess is off by 1 (e.g., dice = 5, guess = 4), show a **neutral face**.

```python
import random

def dice_guessing_game():

   attempts = 0

   dice_value = random.randint(1 , 6) # Generate a random value for the dice roll from 1 to 6

   print(dice_value)

   while True:

       try:

           guess = int(input("Guess the dice value (1-6): "))

           if guess < 1 or guess > 6:

               print("Please enter a number between 1 and 6.")

               continue

           if guess == dice_value:

               print("Correct! You guessed the dice value.")

               print(f"It took you {attempts + 1} attempts to guess the correct value.")

               break

           elif abs(guess - dice_value) == 1:

               print("Close! Your guess is off by 1.")

               attempts += 1

           else:

               print("Incorrect! Better luck next time.")

               attempts += 1
```

```
        except ValueError:
            print("Invalid input. Please enter a number between 1 and 6.")

dice_guessing_game()
```

Output:

```
For testing purpose dice number: 6
Guess the dice value (1-6): 5
Close! Your guess is off by 1.
Guess the dice value (1-6): 7
Please enter a number between 1 and 6.
Guess the dice value (1-6): 1
Incorrect! Better luck next time.
Guess the dice value (1-6): 6
Correct! You guessed the dice value.
It took you 3 attempts to guess the correct value.
```